Expo and Defense

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Expo and Defense before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Improve One Cause

Without opening the old lesson, explain one key idea from Improve One Cause. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Test Normal Cases

Without opening the old lesson, explain one key idea from Test Normal Cases. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Choose Inputs and Outputs

Without opening the old lesson, explain one key idea from Choose **Input**s and **Output**s. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Expo and Defense trace and record the first step you can explain without help.

Use W2: Keep the **final evaluation** record for Evaluating and explaining your project. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Expo and Defense explanation against the supplied facts before discussing it.

Use the worked case for Evaluating and explaining your project. Proposed explanation: “A polished presentation proves the system works”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Judge a different project independently

A fictional library adviser and its manual **baseline** face the same 6 reserved cases. The candidate has 4 full passes; the **baseline** has 5. One additional planned case is unrun by both. No speed data was collected.

The team says “ours is faster and more reliable everywhere” and repairs it after seeing the failures. Explain what the counts support, two unsupported claims and how to label a rerun of the revealed cases.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh **transfer**.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Expo and Defense target to a trace and a justified conclusion.

Defend a sorter with 10/12 routine successes and 2/4 mixed-material successes. Report overall and separate results, show one failure and propose a justified next step. What claim would overstate the **evidence**?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Expo and Defense case and show the deciding evidence.

The expo poster says always correct after six classroom passes.

Write a defensible claim.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

At the expo, a team says our sorter never fails after testing four cards. Replace the claim with an **evidence**-based statement and name a next test.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Expo and Defense adds to Helpful Classroom Sorter and keep the evidence.

Present the final Classroom Sorter portfolio and complete the AI-1 certification **defense**.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Defend a sorter with 10/12 routine successes and 2/4 mixed-material successes. Report overall and separate results, show one failure and propose a justified next step. What claim would overstate the **evidence**?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Record the fixed candidate

Record the fixed candidate. Apply the supplied Expo and Defense facts and keep your reasoning.

Before final cases are revealed, record your version label, where its complete instructions are saved, the agreed **input** and matching rules, the manual **baseline** and any known **limitation**.

Confirm that the instructions stay unchanged while the assigned catalogue is used as **input**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Keep the final evaluation record

> Keep the final evaluation record. Apply the supplied Expo and Defense facts and keep your reasoning.

Use a separate sheet for the teacher's reserved cases. For every case record its ID and **input**s, catalogue and candidate labels, both methods' observed **output**s and reasons, and results after checking. Preserve failure traces.

Here summarise each method's full passes over attempted cases and list unrun cases. State what the comparison supports.

> **My reasoning and evidence**

WORKBOOK · 14 OF 21

W3 Prepare your evidence defence

> Prepare your evidence defence. Apply the supplied Expo and Defense facts and keep your reasoning.

Write a two-minute explanation of the problem, one trace, your actual comparison, a **limitation** and a next action. Explain why this is a written-rule design rather than a model trained on examples.

State which personal information is unnecessary and who decides whether to act.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Judge a different project independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Judge a different project independently. Apply the supplied Expo and Defense facts and keep your reasoning.

A fictional library adviser and its manual **baseline** face the same 6 reserved cases. The candidate has 4 full passes; the **baseline** has 5. One additional planned case is unrun by both. No speed data was collected.

The team says “ours is faster and more reliable everywhere” and repairs it after seeing the failures. Explain what the counts support, two unsupported claims and how to label a rerun of the revealed cases.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Write your final conclusion

> Write your final conclusion. Apply the supplied Expo and Defense facts and keep your reasoning.

Use your own records to complete: Under these conditions, our saved version showed… The **baseline** showed… We still do not know… Our next action is… Include one privacy or human-control choice.

If a case failed, explain its consequence without deleting the result.

> **My reasoning and evidence**

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

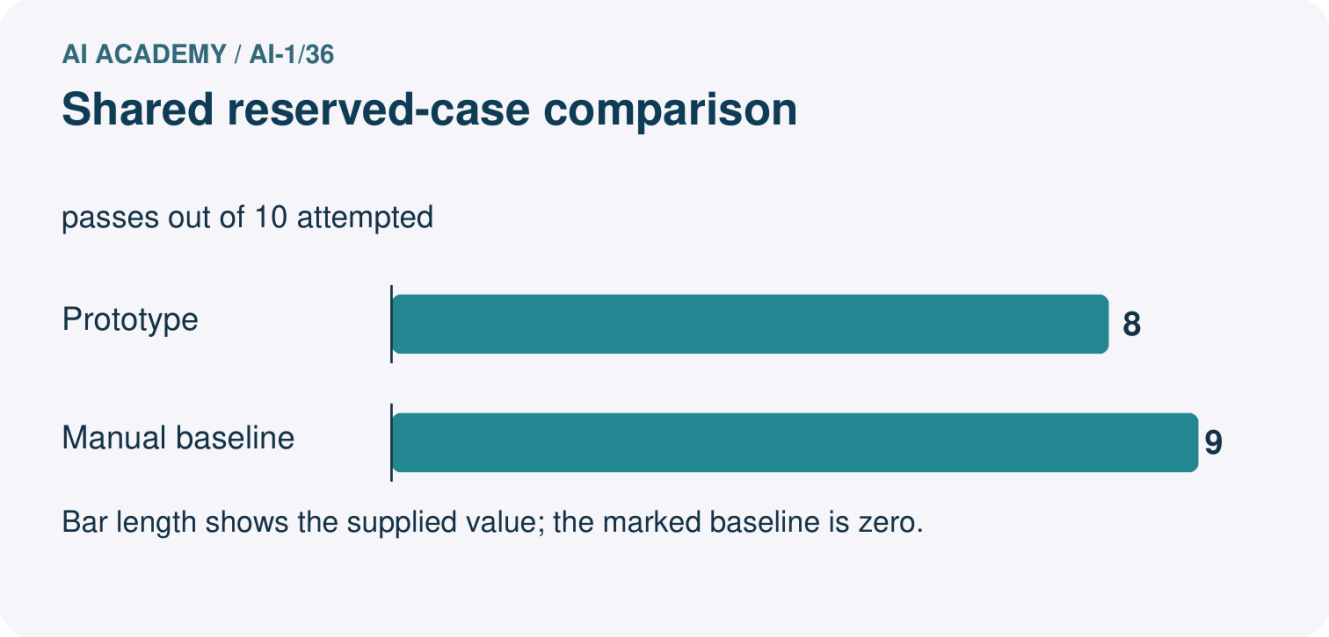
Pictorial case practice: Expo and Defense

> A fictional paper sorter team prepares an exhibition evidence board. Explain the labelled step before checking the changed condition.

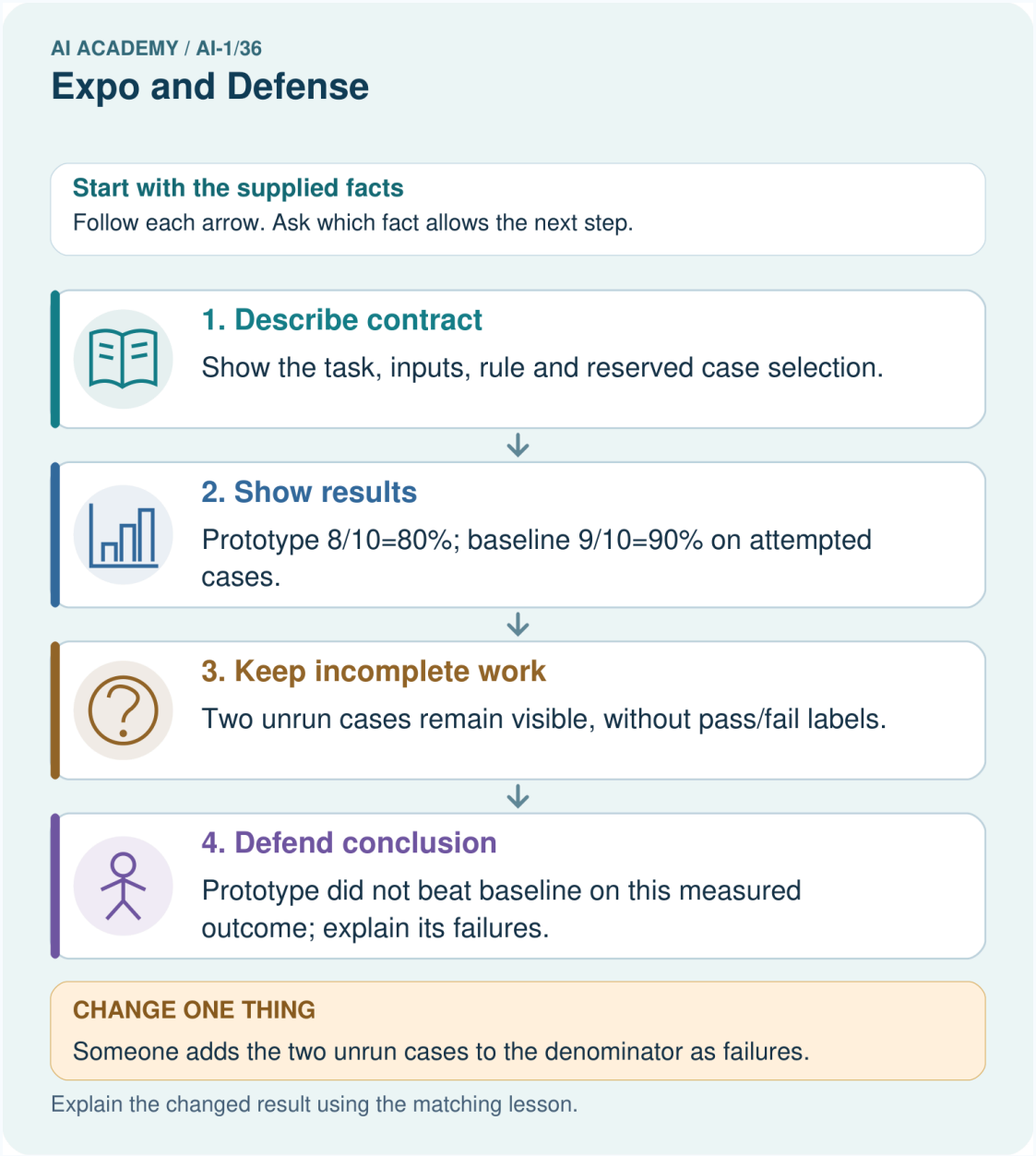
A fictional paper sorter team prepares an exhibition **evidence** board.

Both prototype and manual **baseline** attempt the same 10 reserved cases. Prototype passes 8; **baseline** passes 9.

Two additional planned cases are unrun. No timing or pupil-satisfaction measures were collected.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Write one defensible success statement and one claim the **evidence** cannot support.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Present the Helpful Classroom Sorter's versions, test records, **baseline**, failures and next investigation as an **evidence** portfolio.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Present the Helpful Classroom Sorter's versions, test records, **baseline**, failures and next investigation as an **evidence** portfolio.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional paper sorter team prepares an exhibition **evidence** board.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Prototype did not beat **baseline** on this measured outcome; explain its failures.

1. Keep incomplete work

2. Describe contract

3. Show results

4. Defend conclusion

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Defend conclusion Prototype did not beat **baseline** on this measured outcome; explain its failures. An exhibition should make the **evidence** inspectable.

A polished **demonstration** is not a substitute for the reserved-case record. The shared cases allow a direct descriptive comparison of pass rates; missing timing data prevents a claim that the prototype is faster.

Unrun cases must not be treated as successes.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Someone adds the two unrun cases to the denominator as failures. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. That reports a different measure, 8/12 completed successes out of planned cases.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: That reports a different measure, 8/12 completed successes out of planned cases.

It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

Repair a missing building block

Review Improve One Cause (ai-1/35). Goal: Repair a located defect without changing the agreed requirements. Short practice: Explain why “our repaired version now picks something” is not enough to show success at 6..

Return to this week and explain what you changed.

Review Test Normal Cases (ai-1/33). Goal: Plan routine cases with expected results derived from the requirements. Short practice: Why is none correct for 8 but incorrect for 6? Use both capacity cards..

Return to this week and explain what you changed.

Review Choose **Input**s and **Output**s (ai-1/30). Goal: Turn the project outcome into precise **input** and **output** requirements. Short practice: Trace requests 10 and blank. Why should they not receive the same explanation?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

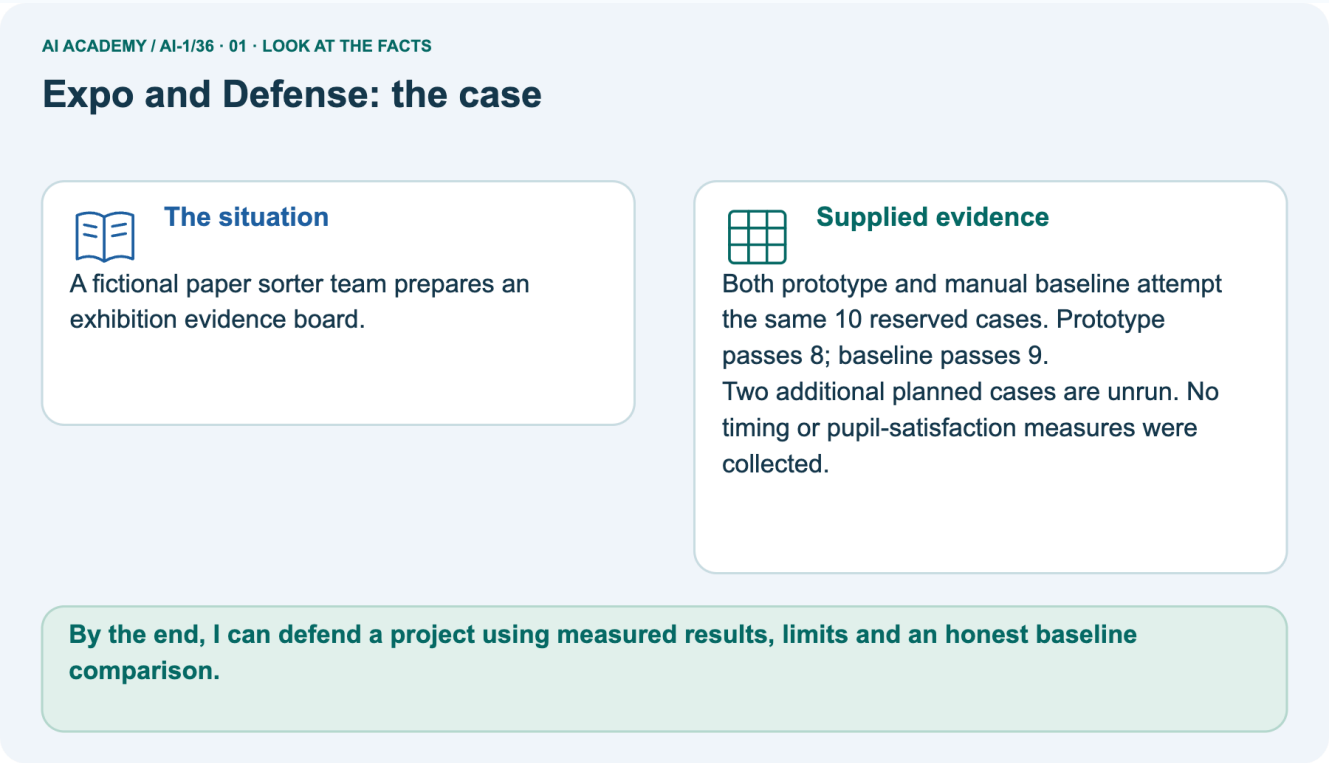
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Expo and Defense picture from its labels, then record a relevant limit.

01. Expo and Defense: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

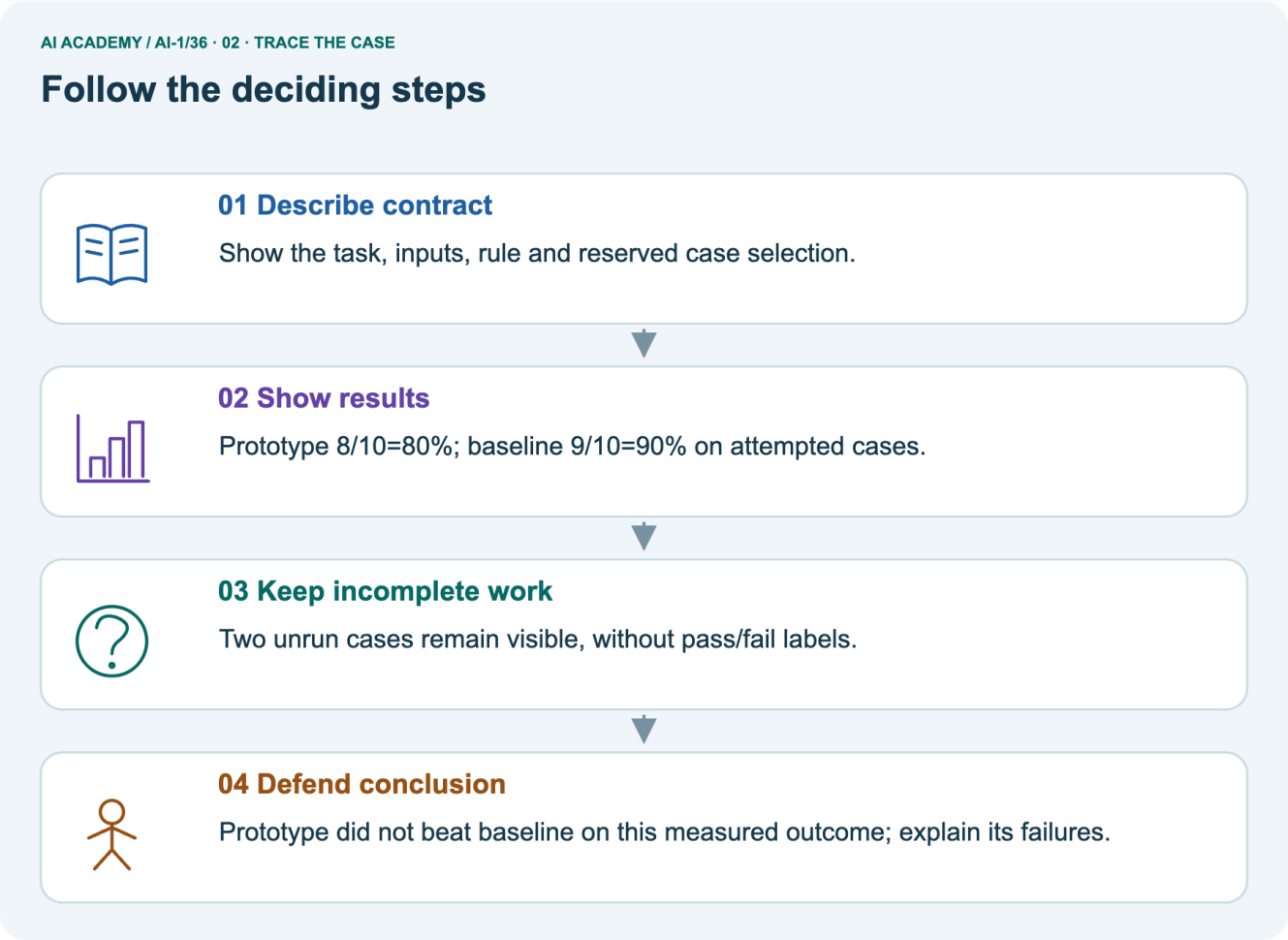
A fictional paper sorter team prepares an exhibition **evidence** board. Supplied facts: Both prototype and manual **baseline** attempt the same 10 reserved cases. Prototype passes 8; **baseline** passes 9.

Two additional planned cases are unrun. No timing or pupil-satisfaction measures were collected.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



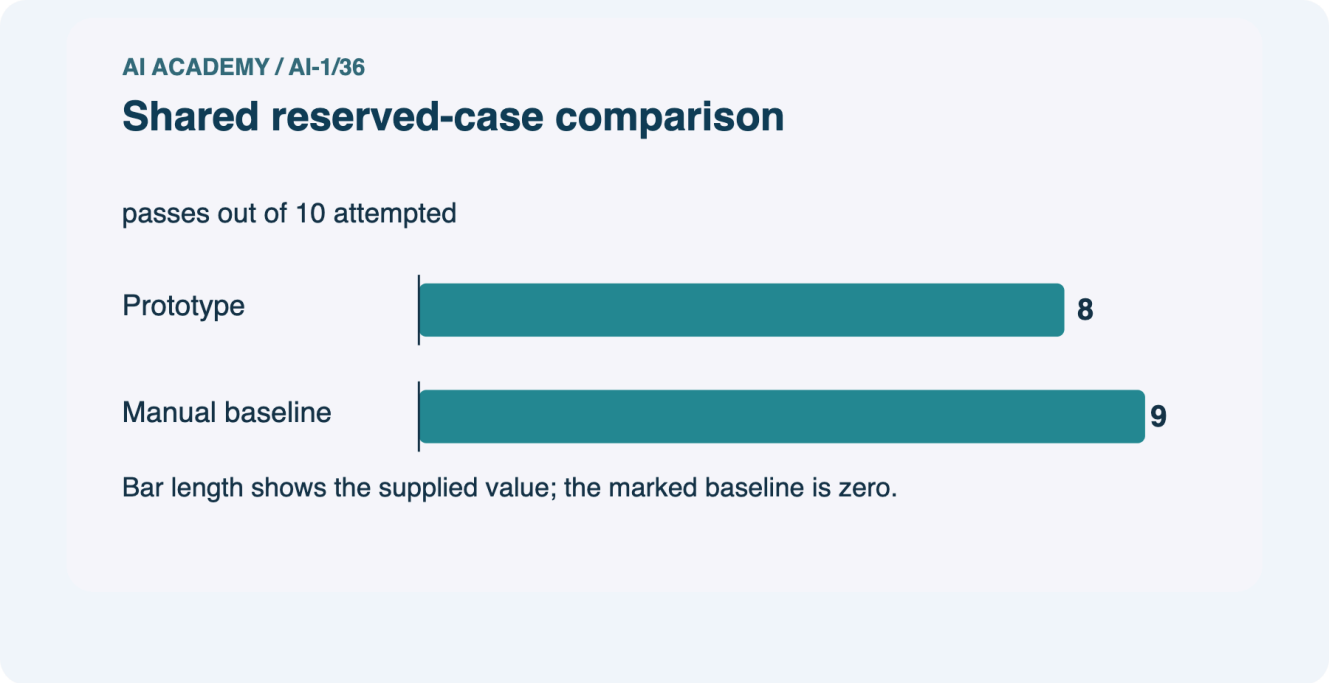
Follow the labelled picture with the matching explanation. Open the vector diagram

Describe contract: Show the task, **input**s, rule and reserved case selection. → Show results: Prototype 8/10=**80%**; **baseline** 9/10=90% on attempted cases. → Keep incomplete work: Two unrun cases remain visible, without pass/fail labels. → Defend conclusion: Prototype did not beat **baseline** on this measured outcome; explain its failures.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Shared reserved-case comparison



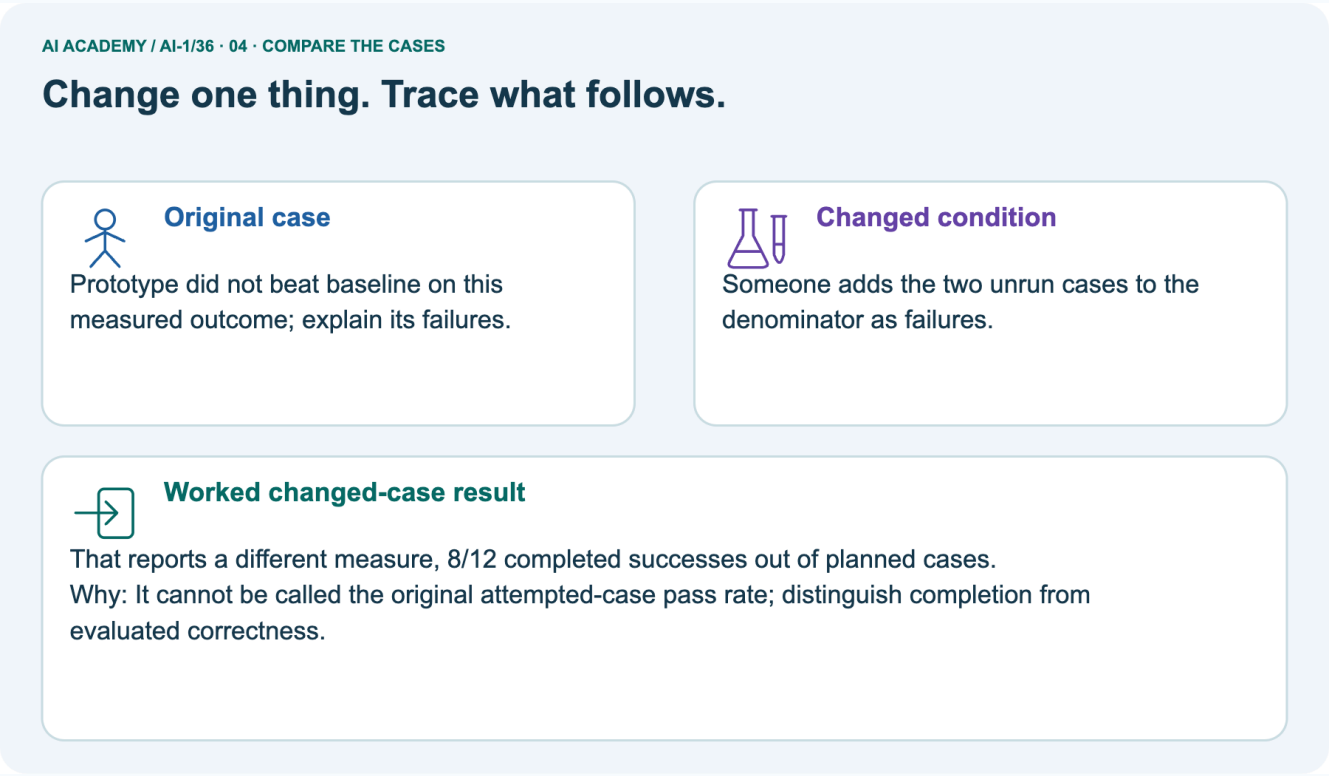
Follow the labelled picture with the matching explanation. Open the vector diagram

Shared reserved-case comparison. Units: passes out of 10 attempted. Prototype: 8. Manual **baseline**: 9.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Prototype did not beat **baseline** on this measured outcome; explain its failures.
Changed condition: Someone adds the two unrun cases to the denominator as failures.

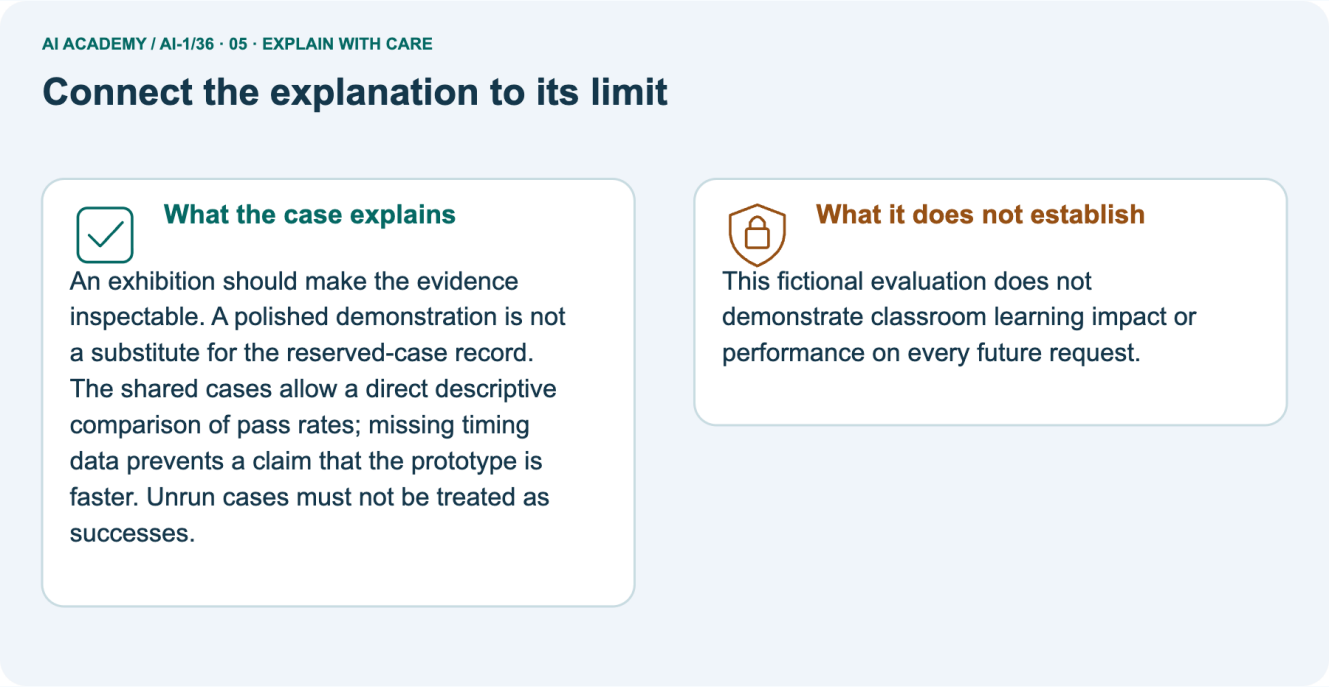
Expected result: That reports a different measure, 8/12 completed successes out of planned cases.
Reason: It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

Predict before checking the worked changed-case result. Name what changed and what the

supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An exhibition should make the **evidence** inspectable. A polished **demonstration** is not a substitute for the reserved-case record.

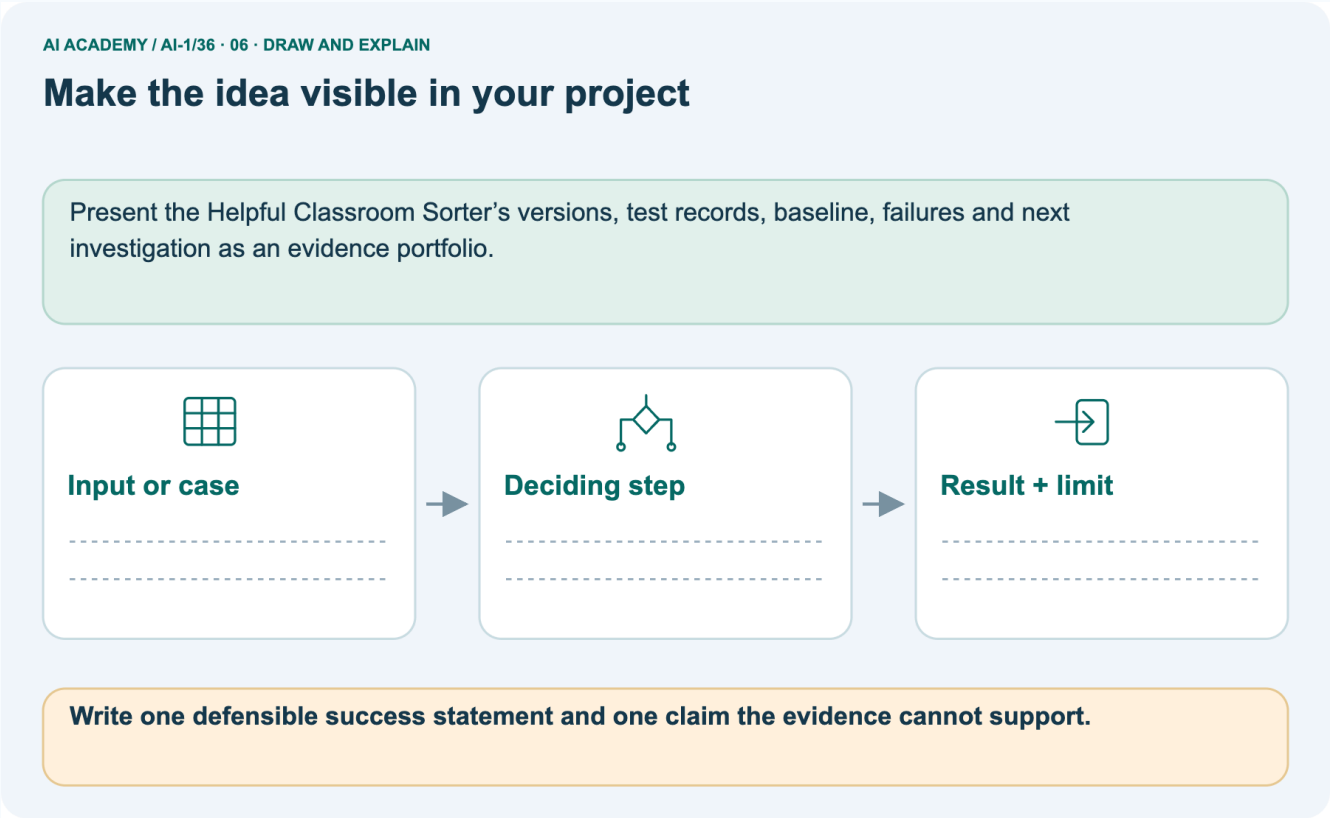
The shared cases allow a direct descriptive comparison of pass rates; missing timing data prevents a claim that the prototype is faster. Unrun cases must not be treated as successes.

Limit: This fictional evaluation does not demonstrate classroom learning impact or performance on every future request.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Present the Helpful Classroom Sorter's versions, test records, **baseline**, failures and next investigation as an **evidence** portfolio. Draw your **input**, deciding step, result and limit.

Supported question: Write one defensible success statement and one claim the **evidence** cannot support.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.